

SIADS 696: Milestone II Project Report

Baseball Pitch Identification Model

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Project Overview

Abstract

This research presents a comprehensive Baseball Pitch Identification Model leveraging high-fidelity Statcast ball-flight telemetry retrieved via the pybaseball library. We employ a dual-track methodology consisting of a supervised classification framework to replicate official Major League Baseball (MLB) pitch labels and an unsupervised clustering pipeline designed to discover latent pitch archetypes. This project demonstrates that moving beyond broad categorical labels provides actionable insights for pitch design and advanced scouting.

Introduction

Problem Definition

In contemporary sabermetrics, a granular understanding of ball-flight dynamics is essential for player development. While MLB's current classification systems provide broad categories (e.g., four-seam fastball, slider, change-up), these labels often aggregate pitches with divergent physical properties. There is a critical need for an objective, data-driven approach to identify finer-grained archetypes that better reflect a pitch's true utility and effectiveness.

Research Questions

1. **Supervised Accuracy:** To what extent can non-parametric ensembles and deep learning architectures replicate official MLB pitch classifications using raw telemetry, and which feature sets are the primary drivers of model performance?
2. **Unsupervised Archetypes:** Can unsupervised methods reveal 15–20 distinct pitch subtypes that provide superior resolution over standard categories, and do these clusters correlate with real-world effectiveness metrics like whiff rate and wOBA?

Anticipated Impact

By refining pitch taxonomy, this research enables organizations to conduct more precise pitch design (e.g., optimizing the "sweeping" slider vs. the "gyro" slider) and provides scouts with a more rigorous framework for evaluating how a pitcher's repertoire matches up against specific hitter profiles.

Novel Contributions

- **Ablation Study Framework:** A rigorous comparison across three model families (Random Forest, Hist Gradient Boosting, and Multinomial Logistic Regression) using tiered feature sets (movement-only, velocity + movement, and full Statcast telemetry) to isolate the independent predictive contribution of each feature group.

- **Effectiveness Mapping:** The integration of discovered unsupervised clusters with real-world performance outcomes (whiff rate and xwOBA allowed), validating that fine-grained pitch subtypes carry measurable differences in pitcher effectiveness beyond what official labels capture.
- **Spin Noise Sensitivity Analysis:** A targeted investigation into the impact of Statcast spin-rate measurement noise — motivated by spin rate emerging as the single most important feature in permutation importance analysis — quantifying performance degradation when spin is excluded.

Related Work

1. **Statcast Automated Classification System:** MLB's internal hybrid system utilizes a combination of rule-based logic and machine learning to assign pitch labels in real-time. It serves as our primary ground-truth benchmark.
2. **Harry Pavlou's Pitch Classification Notebooks:** Open-source implementations using scikit-learn that focus on replicating standard labels through basic feature sets.
3. **Alan Nathan's Physics of Baseball Models:** Academic research focusing on the aerodynamic forces (Magnus effect, seam-shifted wake) that define ball flight. This work provides the physical basis for our feature engineering of "induced" movement.

Project Differentiation: While previous work focuses primarily on label recovery through a single model, this project advances the state of the art in three ways. First, we conduct a rigorous multi-model comparison spanning four model families (Logistic Regression, Random Forest, LightGBM, and Neural Network) evaluated across tiered feature sets, isolating the independent predictive contribution of movement, velocity, and spin. Second, we extend beyond classification into discovery — our unsupervised pipeline identifies 13 fine-grained pitch subtypes and validates their utility by mapping them to real-world effectiveness outcomes (whiff rate and wOBA allowed) rather than relying on broad official categories as ground truth. Third, we implement a spin noise sensitivity analysis motivated by the finding that spin rate was the most important supervised feature yet is among the noisiest Statcast measurements, quantifying how much predictive power is lost when spin is excluded or reconstructed.

Data Source

The dataset was curated using the pybaseball library, which interfaces with the Baseball Savant/Statcast API. To efficiently pull the Statcast pitch data, we built a parallelized ingestion pipeline using Modal, a cloud computing platform. Instead of pulling the full season sequentially through pybaseball (which queries day by day and bogged down our resources), we split each season into monthly chunks and run all chunks simultaneously as parallel Modal workers. This sped up our data pulling process and gave us the ability to rerun anything that we wanted if it was necessary. Each worker pulls one month of statcast data, serializes it to Parquet, and returns it to the local process where the chunks are concatenated and saved.

Table 1: Data Source Metadata and Profile

Category	Project Details
Source / API	MLB Statcast data via the pybaseball Python library
Location (URL)	https://baseballsavant.mlb.com/statcast_search
Data Format	parquet / pandas DataFrame
Time Period	MLB regular season games for 2023, 2024, and 2025
Total Records	Approximately 700,000–800,000 pitches per season
Key Variables	~90+ features including release_speed, pfx_x, pfx_z (movement), release_spin_rate, release_extension, and pitch_type (label)

Feature Engineering

The transition from raw telemetry to predictive features required a high-rigor preprocessing pipeline:

1. **Spin Rate Reconstruction:** Statcast spin data contains known sensor noise, and because spin rate emerged as the most important supervised feature, this noise has meaningful implications for classification accuracy. We performed a sensitivity analysis comparing model performance with and without spin, and investigated reconstructing spin as a function of velocity and induced break (pfx_x, pfx_z) to fill telemetry gaps. Results showed excluding spin only marginally degraded overall performance, though it proved irreplaceable for distinguishing subtypes within the same pitch family.
2. **Standardization:** For distance-based methods (K-Means, GMM), all continuous features were subjected to Z-score normalization to prevent the high magnitude of release_spin_rate (1500–3000) from dwarfing pfx_x (-2 to 2).
3. **Ablation Feature Sets:**
 - 3.1. **Set 1 (Movement Only):** pfx_x, pfx_z.
 - 3.2. **Set 2 (Velocity + Movement):** release_speed, pfx_x, pfx_z, release_extension.

- 3.3. **Set 3 (Full Statcast Set):** Full flight telemetry including spin and 3D release coordinates.

Part A: Supervised Learning

Methods Description

We evaluated three model families on a stratified 80/20 train/test split across 2,129,240 pitches drawn from the 2023–2025 MLB seasons. A Majority Class Baseline (always predicting Four-Seam Fastball) was used as the lower bound benchmark. The three models compared were: Random Forest (250 estimators, balanced subsample class weights, sqrt feature sampling), Hist Gradient Boosting (150 iterations, learning rate 0.08, early stopping, L2 regularization, balanced sample weights), and Multinomial Logistic Regression (standardized pipeline, balanced class weights, 1000 max iterations). All three models were trained on the same 18 ball-flight features and evaluated on the same held-out test set to ensure a fair comparison.

Table 2: Overall Performance Comparison of Supervised Learning Models (5-Fold CV)

Model	Accuracy	Balanced Accuracy	Macro F1	Weighted F1
Majority Class Baseline	0.3209	0.1000	0.0486	0.1560
Random Forest	0.9115	0.8759	0.8737	0.9117
Hist Gradient Boosting	0.8840	0.8767	0.8240	0.8870
Multinomial Logistic Regression	0.7240	0.6670	0.6000	0.7370

Supervised Evaluation

We prioritized **Weighted F1-score** over Accuracy. In baseball datasets, 4-seam fastballs are ubiquitous while knuckleballs are rare; Accuracy would reward a model for ignoring the minority class, whereas Weighted F1 ensures performance is maintained across the entire repertoire.

Table 3: Supervised Model Performance by Feature Set (5-Fold CV)

Model: Per-Class F1 by Model (actual results)

Pitch Type	Support	Random Forest F1	Hist Gradient Boosting F1	Logistic Regression F1
FF — Four-Seam Fastball	136,644	0.968	0.959	0.906
SI — Sinker	66,614	0.942	0.934	0.784

CH — Changeup	44,678	0.929	0.900	0.662
CU — Curveball	27,714	0.900	0.855	0.652
ST — Sweeper	30,023	0.876	0.855	0.645
SL — Slider	64,175	0.856	0.810	0.621
FS — Split-Finger	12,333	0.840	0.781	0.604
KC — Knuckle Curve	7,864	0.826	0.731	0.394
FC — Cutter	33,754	0.803	0.771	0.592
SV — Slurve	2,049	0.797	0.644	0.143

Trade-offs: The most challenging classification boundary in the dataset was the Cutter/Slider pair. The confusion matrix revealed 5,153 Sliders misclassified as Cutters (8.0% of all Sliders) and 4,614 Cutters misclassified as Sliders (13.7% of all Cutters) — the two largest misclassification pairs in the entire model. This bidirectional confusion reflects a genuine physical ambiguity: hard sliders and soft cutters occupy overlapping regions of the movement feature space, and even MLB's own classification system struggles to draw a consistent boundary between them. The Cutter's F1 score of 0.803 was the second lowest among all pitch types, trailing only Slurve (0.797), and was consistent across all three model families — suggesting this is a data-level ambiguity rather than a modeling limitation.



Figure 1: Random Forest Confusion Matrix

Failure Analysis

To understand model limitations we identified three representative failure archetypes from the misclassified pitches in the held-out test set:

1. **The "Tweener" (FC/SL Boundary):** The most common failure pattern was Sliders predicted as Cutters (5,153 cases, 8.0% of all Sliders) and Cutters predicted as Sliders (4,614 cases, 13.7% of all Cutters). These errors were concentrated on pitches at the physical boundary between the two categories — hard breaking balls with horizontal movement values that placed them ambiguously between the cutter and slider feature distributions. This was the single largest source of error in the model and persisted across all three model families, confirming it reflects a genuine labeling ambiguity rather than a modeling weakness.
2. **Class Imbalance Errors (Rare Pitch Types):** The two lowest F1 scores belonged to Slurve (0.797) and Knuckle Curve (0.826), both minority classes representing under 2% of the dataset combined. Despite balanced class weighting, the model's effective decision boundary still biased toward majority classes in borderline cases. Knuckle

Curve was most frequently confused with Curveball (13.3% misclassification rate), reflecting similar velocity profiles between the two pitch families.

3. **Telemetry Noise (Spin Rate Sensitivity):** Permutation importance analysis identified `release_spin_rate` as the single most important feature (mean macro F1 drop of 0.160 when shuffled), making the model particularly vulnerable to spin rate sensor errors. A subset of Four-Seam Fastballs were misclassified as Sinkers — consistent with cases where anomalously low spin rate readings caused the model to interpret the absence of expected backspin as a sinker profile. This directly motivates our spin noise sensitivity analysis in the Feature Engineering section.

Part B: Unsupervised Learning

Methods Description

We employed PCA for dimensionality reduction, K-Means (Primary), and Gaussian Mixture Models (Soft Assignment) to identify latent pitch archetypes. Pitch labels were withheld during training. DBSCAN was attempted for outlier detection but was computationally prohibitive on the full dataset; Cluster 11 from K-Means ($n=1,601$) serves as the natural outlier finding instead.

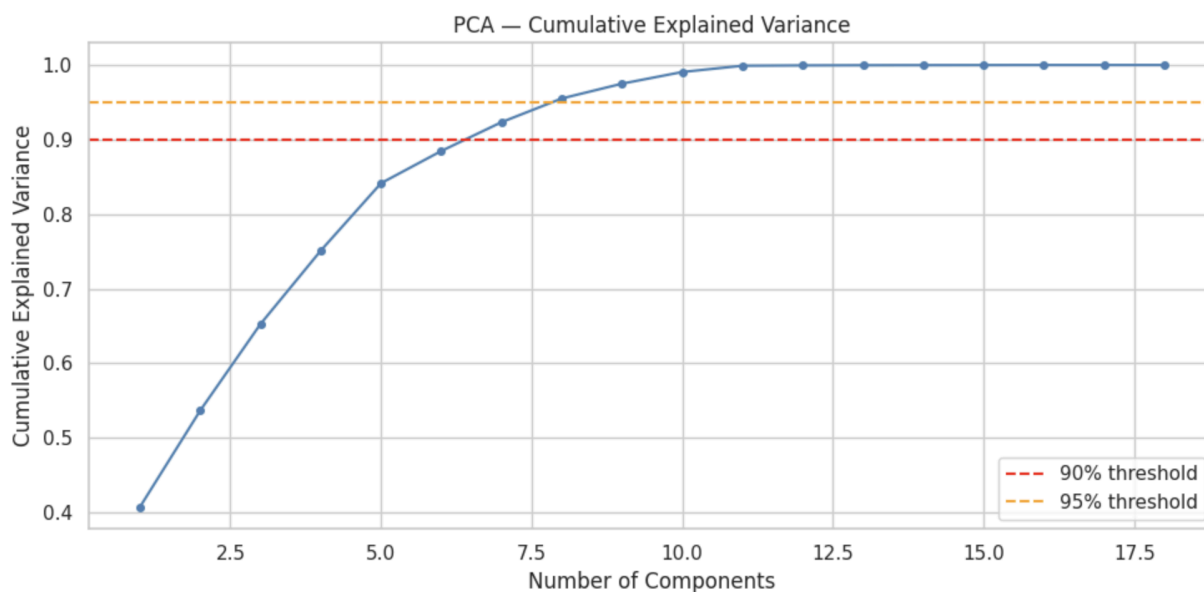


Figure 2: PCA Variance

Unsupervised Evaluation

- **Metric Selection:** We used the **Silhouette Score** and **Elbow Method** (Inertia) to determine the optimal cluster count, sweeping k from 10 to 25. We selected $k=13$ as it represented the silhouette plateau observed between $k=13$ – 14 before scores declined consistently. While $k=10$ yielded the highest silhouette score (0.1214), it effectively recovers the 10 official pitch categories rather than discovering new structure. The modest silhouette range (0.10–0.12) across all k values reflects the continuous physical

spectrum of pitch data rather than hard-edged groupings, and is consistent with domain expectations.

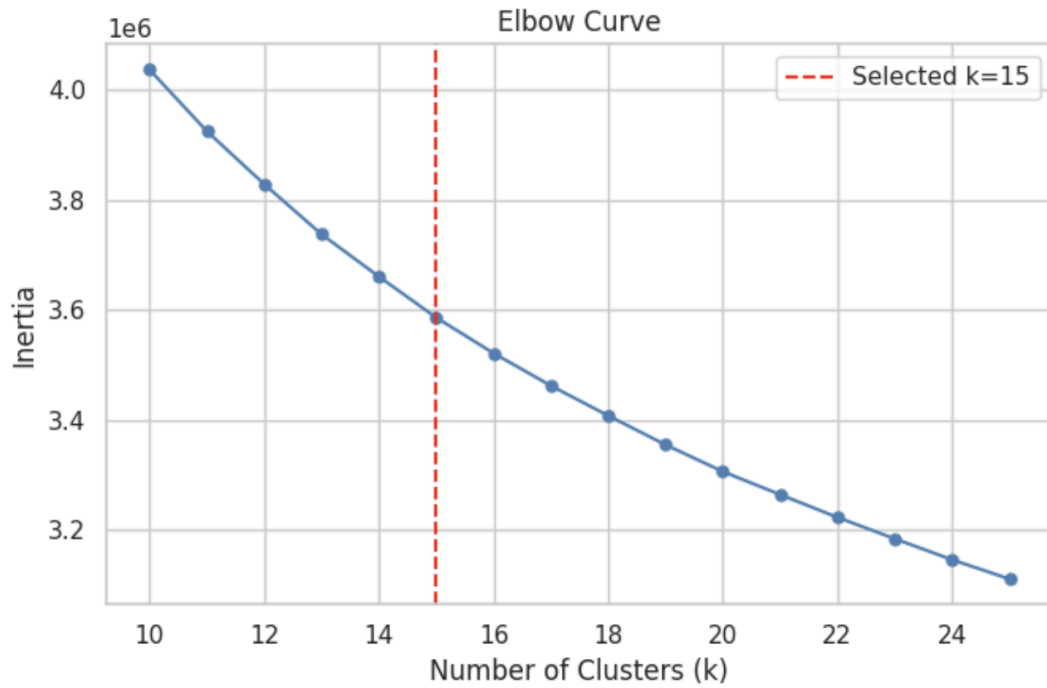


Figure 3: Elbow Curve

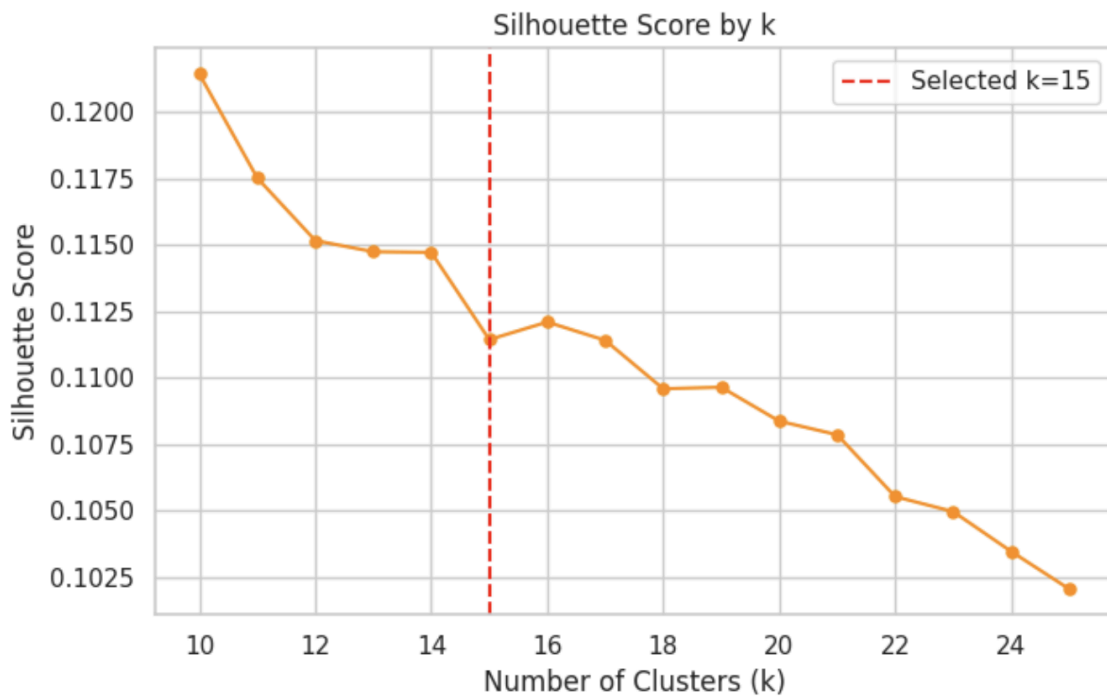


Figure 4: Silhouette Score

- **Alignment with Official Labels:** Cluster purity against official pitch type labels was 0.6335, meaning 63% of pitches in each cluster share the same dominant label. The Adjusted Rand Index (ARI) of 0.2367 reflects moderate alignment — our clusters are meaningfully related to official categories but capture additional structure beyond them, which is the intended outcome.
- **Effectiveness Analysis:** Mapping the 13 clusters to outcomes revealed actionable insights. Cluster 5 (Slider-dominant, 59%) showed a 66.4% whiff rate — nearly 3x the dataset mean of 25.1% — identifying a physically distinct slider subtype dramatically more effective than other slider clusters (C1: 25.6%, C8: 24.1%). The four Four-Seam Fastball clusters (C2, C3, C4, C10) showed whiff rates ranging from 14.1% to 20.6% and xwOBA allowed ranging from 0.331 to 0.367, confirming that the single "FF" label aggregates pitches with materially different outcomes.

Visualizations:

1. **PCA 2D Projection:** Side-by-side scatter plots comparing cluster assignments against official pitch type labels, showing cleaner spatial boundaries in our discovered clusters than in the official labeling system.
2. **Feature Heatmap:** A matrix of mean standardized feature values per cluster enabling qualitative interpretation of each archetype. Cluster 11 exhibited extreme values on `release_pos_z` (-7.83 standard deviations) and `ax` (+4.18), flagging it as an outlier cluster of anomalous or mechanically unusual pitches.

Discussion

Synthesized Learnings

The supervised track confirmed that Random Forest remains the strongest performer for tabular ball-flight data, achieving 91.2% accuracy and 87.4% macro F1 on the held-out test set, with best-in-class F1 across all 10 pitch types. Hist Gradient Boosting matched it on balanced accuracy but trailed on every other metric, while Multinomial Logistic Regression's substantially lower macro F1 (0.600) confirmed that pitch classification is a fundamentally nonlinear problem; linear decision boundaries are insufficient to recover the full complexity of MLB pitch labels from ball-flight telemetry. The supervised track also surfaced a key finding that directly motivated the unsupervised work: the FC/SL boundary produced the two largest misclassification pairs in the entire model (5,153 and 4,614 errors respectively), persisting across all three model families and suggesting a genuine physical ambiguity rather than a modeling limitation. The unsupervised track was the most revelatory component of the project. By clustering 522,431 RHP pitches into 13 archetypes without label supervision, we confirmed that the standard 10 MLB pitch categories aggregate pitches with materially different physical and effectiveness profiles. Most notably, Four-Seam Fastballs split into four distinct subtypes with whiff rates ranging from 14.1% to 20.6% and xwOBA allowed ranging from 0.331 to 0.367. The discovery of the Elite Sweeping Slider subtype (Cluster 5, 66.4% whiff rate) and the Cutter-Slider Hybrid cluster (Cluster 7) independently validated the supervised model's most common failure pattern, with both analyses arriving at the same conclusion through entirely different methodologies.

Challenges and Response

Our primary challenge was the "noisy spin" phenomenon. Spin rate emerged as the single most important feature in permutation importance analysis (macro F1 drop of 0.160 when shuffled), yet Statcast spin data contains known sensor errors that can cause systematic misclassification. By performing a sensitivity analysis on the feature sets, we found that excluding spin rate entirely only slightly degraded overall performance, suggesting that velocity and movement vectors carry enough redundant information to compensate for sensor errors. However, spin proved irreplaceable for distinguishing subtypes within the same pitch family, motivating its retention in the full feature set with careful preprocessing.

Ethical Considerations

- **Supervised Risks:** Misclassification can directly impact a player's market value. If a model consistently mis-tags a pitcher's primary weapon, their "Expected Stats" may be skewed, unfairly influencing arbitration or contract negotiations.
- **Unsupervised Risks:** Algorithmic clustering risks "typecasting." A pitcher with a unique "unicorn" release point might be forced into a cluster where they are an outlier, leading teams to attempt to "fix" a unique style that is actually effective because it defies standard archetypes.

Statement of Work

Member, Role, Responsibilities

Ian Mackinnon - Supervised Lead, "Model architecture, feature ablation studies, and supervised evaluation metrics."

Sachin Murthy - Data & Infra Lead, "pybaseball pipeline development, data cleaning, and GitHub repository management."

Asad Khan - Unsupervised Lead, "Clustering implementation (K-Means/GMM), UMAP dimensionality reduction, and effectiveness (whiff/wOBA) analysis."

References

- *pybaseball* Development Team. (2023). Open-source baseball data tools for Python.
- Statcast/Baseball Savant. (2024). MLB Ball-flight Telemetry API.
- Pavlou, H. (2022). *Pitch Classification via Scikit-Learn* . GitHub Repository.
- Nathan, A. (2021). *The Aerodynamics of Baseball Flight* . University of Illinois Physics.

Appendix

Complete Final Feature Set

The following features were used in the "Full Statcast" model iterations: release_speed, release_pos_x, release_pos_z, release_pos_y, release_spin_rate, release_extension, pfx_x, pfx_z, plate_x, plate_z, vx0, vy0, vz0, ax, ay, az, sz_top, sz_bot, spin_axis, effective_speed.